

Revisiting IPS-based Algorithms for Off-Policy Evaluation of Contextual Bandits

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Abstract

Off-policy evaluation (OPE) is widely used to compare contextual bandit policies in recommender systems. While there a lot of recent methodological developments, suggesting novel OPE schemes, they are typically validated in the synthetic environments, which not necessarily possess the structure of the real-world datasets. In this paper, we consider the inverse propensity score (IPS) method and its modifications, and study how empirical conclusions inferred from the data depend on evaluation pipelines. We show, that even in the synthetic environments, rankings of different estimators are sensitive to random seeds, log generators, and sample size. Using the popular benchmark, the Open Bandit Dataset [11], we analyze logging behavior and data characteristics that may violate the i.i.d. assumptions of the log generation.

Code — <https://github.com/DariaK05/opec-eval-2026/>.

CCS Concepts

• **Information systems** → **Evaluation of retrieval results; Recommender systems**; • **Computing methodologies** → *Simulation evaluation*; • **General and reference** → Empirical studies.

Keywords

Contextual bandits, Off-Policy Evaluation

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1 Problem setup

Off-Policy Evaluation (OPE) aims to evaluate the effectiveness of new policies based on the logged data collected by another, behavioral policy. Typically the methods of this family use the Inverse

Propensity Scoring (IPS) [10, 14] or its modifications. However, this method and its modifications are known to suffer from high variance and sensitivity to rare actions, especially when the action space is large [12], despite all efforts to create more stable versions of these algorithms [2, 7, 8] and develop more robust versions of evaluation pipelines [13]. In this work, we revisit common OPE estimators and demonstrate that evaluating algorithms on synthetic benchmarks can yield inconsistent results, while the only available real-world dataset for OPE evaluation has several issues, which we highlight and analyze empirically.

Contextual bandit setting. We briefly describe the general pipeline of contextual bandits. Let $x \in \mathcal{X} \subseteq \mathbb{R}^{d_x}$ denote a d_x -dimensional context vector drawn i.i.d. from an unknown distribution $p(x)$. Given a context x , an action a is selected according to a policy $\pi(a | x)$ from a finite action space $\mathcal{A}$. The corresponding reward $r \in [0, r_{\max}]$ is then sampled from an unknown conditional distribution $p(r | x, a)$. The value of a policy π is defined as

$$V(\pi) := \mathbb{E}_{p(x)\pi(a|x)p(r|x,a)}[r] = \mathbb{E}_{p(x)\pi(a|x)}[q(x,a)], \quad (1)$$

where $q(x,a) := \mathbb{E}_{p(r|x,a)}[r]$ denotes the expected reward given context x and action a . OPE aims to develop an estimator $\hat{V}$ given the access to a fixed dataset $\mathcal{D} = \{x_i, a_i, r_i\}_{i=1}^n$ of n independent interactions logged with some sequence of policies $\pi_{0,i}$. Formally, $(x_i, a_i, r_i) \sim p(x_i)\pi_{0,i}(a_i|x_i)p(r_i|x_i, a_i)$. There are two common data-collection settings:

- (1) Fixed behavioral policy π_0 not depending upon i ;
- (2) Time-varying behavioral policies ($\pi_{0,i}$ depends on index i).

In this paper, we discuss and cover a number of classical and more recent OPE algorithms: the Direct Method (DM), Inverse Propensity Scoring (IPS), Self-Normalized Inverse Propensity Scoring (SNIPS), Doubly Robust (DR), Marginalized IPS (MIPS, [12]) and Context-Huddling Inverse Propensity Score (CHIPS, [5]).

Direct Method (DM). The DM method first fits a reward model $\hat{q}(x,a)$ by regression on the logged data $\mathcal{D}$ and estimates $V(\pi)$ with

$$\hat{V}_{\text{DM}}(\pi; \mathcal{D}, \hat{q}) := \frac{1}{n} \sum_{i=1}^n \sum_{a \in \mathcal{A}} \pi(a|x_i) \hat{q}(x_i, a). \quad (2)$$

DM typically has low variance but can be biased, if the regression estimator $\hat{q}$ is not properly specified [2].

Inverse Propensity Scoring (IPS). The IPS estimator [10] is an approach based on the importance sampling paradigm [9]:

$$\hat{V}_{\text{IPS}} = \frac{1}{n} \sum_{i=1}^n \frac{\pi(a_i | x_i)}{\pi_{0,i}(a_i | x_i)} r_i = \frac{1}{n} \sum_{i=1}^n w_i(x_i, a_i) r_i, \quad (3)$$

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where $w_i(x, a) = \pi(a | x) / \pi_{0,i}(a | x)$ is called the importance weight. IPS is unbiased if $\pi(\cdot | x) \ll \pi_{0,i}(\cdot | x)$ for all $x \in \mathcal{X}$ and any i . However, importance weights $w_i(x_i, a_i)$ can take large values, especially if the action space is large [12], leading to high variance of $\hat{V}_{\text{IPS}}$ and numerical instability.

Self-Normalized IPS (SNIPS). The SNIPS estimator [15] attempts to reduce IPS variance with importance weights normalization:

$$\hat{V}_{\text{SNIPS}} = \frac{\sum_{i=1}^n \frac{\pi(a_i | x_i)}{\pi_{0,i}(a_i | x_i)} r_i}{\sum_{i=1}^n \frac{\pi(a_i | x_i)}{\pi_{0,i}(a_i | x_i)}} = \frac{\hat{V}_{\text{IPS}}}{\hat{V}_{\text{SN}}}. \quad (4)$$

It introduces additional bias, yet is capable of reducing variance and the overall MSE.

Doubly Robust (DR). The DR estimator [2] reduces the variance of IPS by using the regression-based estimator from DM:

$$\hat{V}_{\text{DR}} = \hat{V}_{\text{DM}}(\pi; \mathcal{D}, \hat{q}) + \frac{1}{n} \sum_{i=1}^n \frac{\pi(a_i | x_i)}{\pi_{0,i}(a_i | x_i)} (r_i - \hat{q}(x_i, a_i)). \quad (5)$$

DR is unbiased if either the reward model $\hat{q}$ or the logging policies $\pi_{0,i}$ is correctly specified (see e.g. [2]).

Marginalized IPS (MIPS) [12]. MIPS algorithm aims to reduce the IPS variance by introducing action embeddings. Formally, we consider a random variable (embedding) $e \in \mathbb{R}^{d_e}$ drawn from some possibly unknown distribution $p(e | x, a)$. Then MIPS estimator is defined as

$$\hat{V}_{\text{MIPS}}(\pi, \mathcal{D}) := \frac{1}{n} \sum_{i=1}^n \frac{p(e_i | x_i, \pi)}{p(e_i | x_i, \pi_{0,i})} r_i := \frac{1}{n} \sum_{i=1}^n \tilde{w}_i(x_i, e_i) r_i, \quad (6)$$

where $p(e | x, \pi) = \sum_{a \in \mathcal{A}} p(e | x, a) \pi(a | x)$. MIPS estimator is unbiased under particular conditions, and has lower variance than IPS, see [12]. Yet to deal with the typically unknown distribution $p(e | x, a)$, [12] suggests to estimate $\tilde{w}_i(x_i, e_i) = \mathbb{E}_{\hat{\pi}_{0,e,i}(a | x_i, e_i)} w_i(x_i, a)$, where $\hat{\pi}_{0,e,i}(a | x, e)$ can be estimated by regressing on x, e . This step can be highly unreliable with small amounts of data.

Context-Huddling Inverse Propensity Score (CHIPS), [5]. In [5], the authors assume that some *context* clustering function $\phi(x) : \mathcal{X} \rightarrow \mathcal{C}$ is given and set $p(a | c, \pi) = \int_{\mathcal{X}} \pi(a | x) p(x | c) dx$. Then the CHIPS estimator is defined as

$$\hat{V}_e^{\text{CHIPS}} = \frac{1}{n} \sum_{i=1}^n \frac{p(a_i | c_i, \pi)}{p(a_i | c_i, \pi_{0,i})} r_i. \quad (7)$$

CHIPS estimator is unbiased under conditions similar to the ones required by MIPS, yet estimating $p(a | c, \pi_{0,i})$ from logged data requires either to know the *full* distribution $\pi_{0,i}(a | x)$ for all x in the logged data or sufficiently many samples from each $\pi_{0,i}$, which are typically not available.

In summary, classical methods (IPS, DR, DM) remain valid under both fixed and time-varying behavioral policies. In contrast, it is unclear, if advanced OPE methods, such as CHIPS and MIPS can be applied when the behavioral policy is not stationary in real scenarios.

2 Synthetic data evaluation

In this section, we conduct experiments on a synthetic dataset to assess the stability and consistency of off-policy evaluation methods.

Dependency on the particular data log. In order to study the sensitivity of the quality of OPE methods to the particular data log, we adopt the synthetic data generation procedure introduced in [12]. In this setup, context vectors $\{x_i\}_{i=1}^n$ are sampled from certain density $p(x)$, and each discrete action a is associated with a latent embedding e drawn from a conditional distribution $p(e | a)$. The expected reward $q(x, e)$ is then synthesized as a function of context, embedding, and additional randomly sampled parameters. The behavioral policy π_0 is fixed and given by applying the softmax function to $q(x, a) = \mathbb{E}_{p(e|a)} [q(x, e)]$. We reproduce the experimental configuration of Saito et al. [12], varying only the random state that controls the generation of the synthetic log, size of logged dataset and number of actions. Additionally, we generate a separate test dataset to compute the ground-truth policy value. The mean squared error (MSE) is calculated between the estimated and ground-truth policy values. We evaluate IPS, SNIPS, DM, DR, CHIPS (with 5 context clusters), MIPS and MIPS (true) (which is MIPS version with true importance weights obtained in the process of generating synthetic data).

Comparing OPE estimators on different logs generated from the same environment. We analyze how the sample size of the synthetic log affects the ranking of off-policy evaluation methods under different instances of environments. Precisely, we fix the expected reward function $q(x, e)$, and thus the behavioral policy $\pi_0(a | x)$. We first show the MSE plots for two instances of environment, corresponding to $N_A = 500$ actions, and different reward designs $q(x, e)$ in Figure 1 and Figure 2, respectively. On both plots, the ranking of the estimators changes as the sample size n increases. What is perhaps more important is that the ranking of different methods varies for different reward designs (only varying random seeds for coefficient generation).

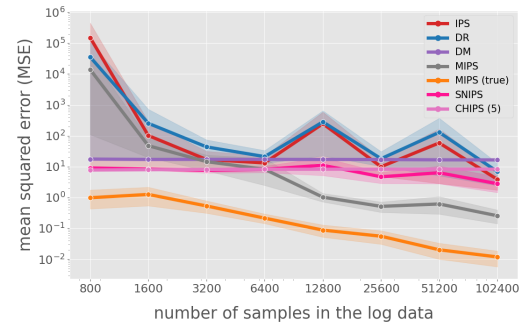


Figure 1: MSE of various OPE methods, *random_state* = 5.

Testing policy ranking consistency. To quantify the consistency of the ranking of estimates, we reran OPE on 25 validation datasets from the same environment for 500 actions and 102400 samples and calculated Spearman's rank correlations between the MSE-based ranking of the six estimates (excluding MIPS(true), as it is not applicable to real data). We averaged the final matrix over 2 different environment configurations. The results in Table 1 show that while two pairs of estimates (IPS and DR; IPS and SNIPS) have relatively high correlation, others, on the contrary, show almost no correlation. This indicates that the rankings of these OPE methods instantly swap over different particular datasets from the same environment.

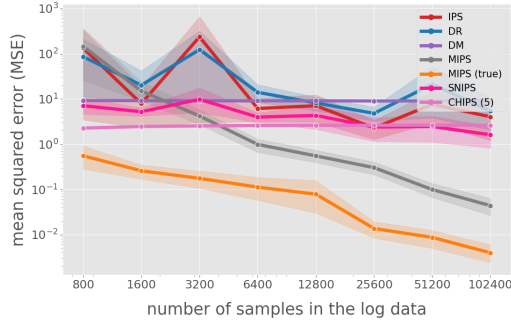


Figure 2: MSE of various OPE methods, $random_state = 12345$.

Table 1: Spearman’s rank correlation for OPE estimators.

	IPS	DM	DR	MIPS	CHIPS (5)	SNIPS
IPS	1.000	-0.264	0.851	0.019	-0.140	0.668
DM	-0.264	1.000	-0.155	0.127	0.025	-0.074
DR	0.851	-0.155	1.000	0.033	-0.031	0.496
MIPS	0.019	0.127	0.033	1.000	0.030	-0.039
CHIPS (5)	-0.140	0.025	-0.031	0.030	1.000	-0.044
SNIPS	0.668	-0.074	0.496	-0.039	-0.044	1.000

These results suggest that making conclusions from synthetic experiments may strongly depend on the particular data realization and the chosen sample size. While huge variance of OPE estimates was already described in literature, available methodology for robust OPE evaluation [13] suggests to use some statistics inferred from multiple run of OPE methods (such as respective CDFs) in a fixed particular environment.

Varying the sample size and random state. For every fixed sample size, we consider 25 different configurations of the synthetic environment with 500 actions. For each of them we create a test dataset to obtain ground-truth policy value and a validation dataset, on which we perform OPE to then calculate MSE for all methods using the corresponding ground-truth. Final MSE for each estimator is calculated via averaging over 25 configurations results. The results are reported in Figure 3, showing potentially more stable behavior compared to the plots reported for a fixed environments in Figures 1 and 2.

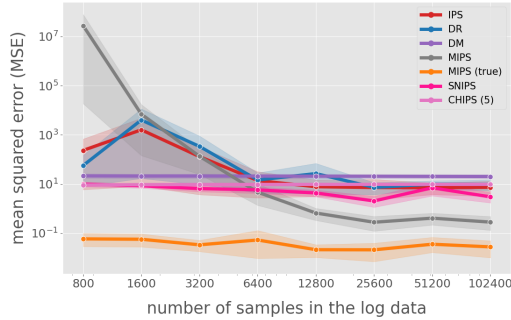


Figure 3: MSE of various OPE methods for varying numbers of sample sizes over 25 different environments.

In one of the rare attempts to mitigate the issues described above, [3] proposed AutoOPE: an ML-based method, which aims to predict the MSE of the particular estimator on the particular data, thus accounting for the sensitivity to the particular log. However, considering the variety of different environments and estimators, reliability of such estimator selection or estimator robustness evaluation approaches is highly questionable.

To sum up, the synthetic experiments outlined in this section show that the relative performance rankings of OPE estimators are highly sensitive to the particular generated log and to the sample size, even when the underlying environment is fixed. Thus the conclusions drawn from a particular synthetic benchmark may be misleading, which motivates a closer study of OPE estimators on the real data in the next section.

3 Real data evaluation

Real-world open datasets for OPE research are limited, and existing alternatives often require substantial modifications that make them less representative compared to the real-world data. To the best of our knowledge, the Open Bandit Dataset [11] is the only dataset collected with a logged policy. Many recent OPE studies [3, 5, 12, 13] rely on this particular dataset.

Open Bandit Dataset (OBD, [11]). The OBD is a large-scale logged contextual bandit dataset collected from the ZOZOTOWN e-commerce platform. It contains two logging policies: Random, with approximately 1.3×10^6 interactions, and Bernoulli Thompson Sampling (BTS), with 1.2×10^7 interactions. Each interaction records the user context, item context, the chosen item from 1 to 80 and its position from 1 to 3 (together forming 240 actions), the probability of an item being recommended at the given position (propensity score), and the observed reward (click).

Policy and environment consistency. A careful inspection of the Open Bandit Dataset with BTS policy shows that the logging policy changed during the experiment. We illustrate this in Figure 4, where we plot the histogram of propensity scores for (action, position) pair with the largest standard deviation across the logged propensity scores. Moreover, [11] does not provide information about the frequency or timing of these updates. As a result, advanced OPE methods such as MIPS and CHIPS cannot be reliably applied to the dataset with BTS policy. This is the reason why, as logged data, researchers typically use the part of the Open Bandit Dataset collected under the Random policy. At the same time, this pipeline still contains several inaccuracies.

Biased ground-truth computations. In many studies (see e.g. [3, 12]), off-policy evaluation is conducted in the following pipeline: the dataset with Random policy is used as logged data, while the BTS policy, with the prior corresponding to the beginning of the ZOZOTOWN experiment, is treated as the target policy. The ground-truth value of this target policy is then estimated using the dataset with BTS policy. At the same time, given the time-varying behavior policy logged in the BTS dataset, it can not be used for the ground-truth evaluations without introducing an additional bias, which is hard to quantify. We are not aware of any estimates of this bias in the literature.

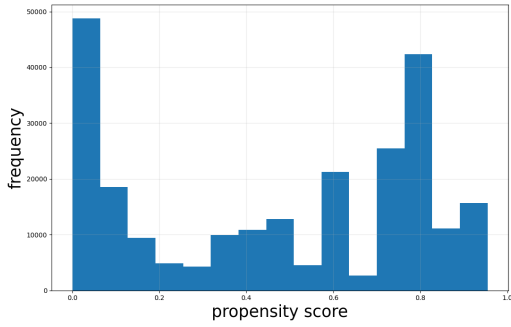


Figure 4: Propensity scores distribution for (item, position) with the highest standard deviation from OBD BTS ALL log.

Inconsistencies in the Random part of the OBD. There are two common ways of using the Random-policy part of the Open Bandit Dataset as logged data.

Several papers [4, 13] treat the logged policy as a uniform one over 80 actions, ignoring item's position. In this case, off-policy evaluation is performed with target policy $\pi(a)$ and logged policy $\pi_0(a|\text{position})$ and then averaged over all interactions. This hypothesis implies, in particular, that the $\pi_0(a|\text{position})$ distribution of action probabilities given position is uniform with probability of each action choice $1/80$. To verify this assumption, we conducted a simple Pearson χ^2 test based on the data, collected from the Random subset of OBD. As shown in Table 2, this hypothesis is rejected. This raises the question, if the logged probability value $1/80$ can be used for computing propensity scores.

Second group of papers [1, 5, 12] treat the logged policy as a uniform one over 240 actions, identifying an action with a pair (item, position). We conducted a χ^2 test for the hypothesis that the distribution over (item, position) pairs is uniform. As shown again in Table 2, the test rejects this hypothesis, too. Thus, neither of the commonly used assumptions about logged probabilities are supported by statistical evidence and can potentially lead to misspecified propensity scores.

Table 2: Results of chi-square tests on OBD UR ALL log. Item conditional distribution considered for position 1.

Test	χ^2 statistic	DoF	p-value
Item-Position uniformity	2186.15	239	$< 10^{-8}$
Item uniformity	752.18	79	$< 10^{-8}$
Item conditional uniformity	749.38	79	$< 10^{-8}$

Absence of user id. The OBD does not include user identifiers, which limits the types of policies and off-policy methods that can be used. Many methods in contextual bandits and recommender systems [6] require distinguishing individual users or constructing user-level representations, which is not possible in this setting. Indeed, the OBD dataset contains only 4 user features, resulting in few unique feature combinations in the "all" campaign. As a result, multiple users share the same feature vector.

4 Conclusion

We have shown several issues related with OPE evaluation protocols. First, choice of the "best" estimator based on synthetic dataset

can be extremely unstable, depending strongly on the specific log realization and sample size. On the only widely used real-world benchmark, the OBD, we further documented several peculiarities (non-stationarity of the BTS logging policy, deviations of the Random log from commonly assumed uniform propensities over actions), which complicate fair evaluation and make it impossible to apply many novel variance-reduced OPE methods. Together, these findings call for more careful reporting and scrutiny of experimental setups. They also highlight the need for evaluation pipelines that explicitly account for non-stationary logging and uncertainty in logged propensities. In addition, they emphasize the importance of more clearly documented OPE datasets in order to support reliable benchmarking of OPE methods.

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